Preinjury and Postinjury Factors That Predict Sports-Related Concussion and Clinical Recovery Time

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Abstract

Objectives: To evaluate the relationship between preinjury risk factors (RFs) and subsequent occurrence of concussion and examine whether preinjury RFs or postinjury assessments predict clinical recovery in collegiate athletes. **Methods:** Risk factors (sex, sport, and self-report history of concussion, migraine, attention-deficit disorder, learning disability, depression, and anxiety) and Sport Concussion Assessment Tool (SCAT), depression/anxiety screenings, and neuropsychological testing were obtained before the season. For athletes who sustained concussion, RFs, postinjury SCAT, neuropsychological assessment, and clinical recovery were assessed. **Results:** We assessed 1152 athletes (69% male) at baseline and 145 (75% male) after subsequent concussion diagnosis. Only sport type (Wald = 40.29, $P = 0.007$) and concussion history (Wald = 9.91, $P = 0.007$) accounted for unique variance in subsequent concussion. Of athletes followed until full recovery, mean days until symptom-free (DUSF) was 9.84 ± 11.11 days (n = 138, median = 5 days, range = 1-86) and mean days until full return to play (DUFRTP) was 20.21 ± 19.17 (n = 98, median = 20.21, range = 4-150). None of the RFs or baseline testing measures were associated with DUSF or DUFRTP (P's > 0.05). After injury, athletes who reported more total symptoms ($r_s = 0.31$, $P < 0.001$) and higher symptom severity ($r_s = 0.33$, $P < 0.001$) exhibited longer DUSF. Days until symptom-free correlated with DUFRTP ($r_s = 0.75$, $P < 0.001$). Among athletes assessed within 2 days after injury, DUSF was associated with Immediate Postconcussion Assessment and Cognitive Test visual motor ($r_s = -0.31$, $P = 0.004$), reaction time ($r_s = 0.40$, $P < 0.001$), and symptom score ($r_s = 0.54$, $P < 0.001$). **Conclusions:** Only sport type and concussion history predicted subsequent occurrence of concussion, and none of the RFs or baseline measures predicted clinical recovery. Immediate postinjury assessments, including symptom number and severity, and select clinical measures predicted longer clinical recovery.

Key Words: modifiers, sex, neurocognitive evaluation, SCAT, symptoms, return to play

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BACKGROUND

Sport-related concussion (SRC) is one of the top 3 time-loss injuries for high school and college athletes.[1] Most SRC research has focused on male American football, with less attention being focused on women or athletes playing other sports (eg, soccer, lacrosse, ice hockey, and rugby), although the combined rate of concussion across these other sports is also high, underscoring the importance of studying SRC across a variety of sports.[1–3] Most athletes experience symptom recovery within 2 weeks of SRC, although a small percentage experience protracted recovery, with symptoms lasting 21 days or longer.[4,6–11] Understanding the factors that increase an athletes' susceptibility to SRC, as well as predicting those at risk for prolonged recovery, is of particular importance to clinicians, aiding them in providing athletes with realistic information regarding their risk for injury, as well as research-informed timelines of when they are likely to return to learn and return to play (RTP) after injury. For example, it has been demonstrated that early psychoeducational interventions promote a better understanding of the nature and course of concussion, help to "normalize" symptoms, decrease misattribution of symptoms, minimize the emergence of psychological symptoms (eg, anxiety and depression) and allow patients to set appropriate expectations for recovery.[12,13]

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Risk Factors

A challenge for clinicians managing SRC is determining whether there are risk factors (RFs) that predict the occurrence of and outcome after injury. The research related to these factors has used the term "modifiers"[9]; however, we instead will use the term "RFs" because it is more descriptive and consistent with the broader sports medicine literature. Risk factors can be defined as those factors that may be associated with, or influence, clinical recovery from SRC.[9,10] Preinjury RFs include history of concussion,[14–16] age,[17–19] depression, or other mental health difficulties,[20–22] headache disorder/migraines,[3] learning disability (LD), attention-deficit hyperactivity disorder (ADHD),[23,24]

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and female sex.[17,24–26] After injury RFs include those listed above as preinjury RFs, as well as the characteristics of the injury itself, specifically the signs [eg, loss of consciousness (LOC) and amnesia], symptoms (eg, headache and dizziness), and performance on other clinical assessments (eg, tests of cognitive function and balance).

Research regarding RFs for prolonged recovery from SRC has been the topic of systematic reviews,[9,10] which note that little prospective research evaluating whether these RFs are related to the length of recovery after SRC. One of the most robust findings to date is that symptom severity within 24 hours of an injury is predictive of clinical recovery time.[4,10,27,29–31] The development of headaches or migraines after SRC[27,28] and depression after SRC has been associated with increased clinical recovery time.[21,22] Whether there are specific acute postinjury symptoms or signs that predict prolonged recovery remains controversial.[10] Some studies have reported that LOC, posttraumatic amnesia, retrograde amnesia, or the presence of dizziness and headache at the time of concussion may be associated with prolonged recovery.[6,29] Additional research has also identified poor performance on Immediate Postconcussion Assessment and Cognitive Test (ImPACT) visual memory and reaction time composites as predicting prolonged recovery.[29]

Assessment tools have also been used to predict SRC recovery with the current standard for concussion management being a multimodal assessment, such as the Sport Concussion Assessment Tool (SCAT),[32,33] during the acute/diagnostic phase of the injury. The Concussion in Sport Group (CISG) "Berlin" statement[7] underscored that the SCAT is most useful in the acute injury phase, especially within the first 3 days after SRC. We have previously demonstrated the utility of the SCAT2[34] in acutely assessing SRC; the subsequent versions of the SCAT, the SCAT3[34] and SCAT5,[32,33] include the same key components of the SCAT2. Performance on these measures may be used to further understand the role of postinjury RFs in clinical recovery because they are typically integrated with clinical data to make RTP decisions. In addition, neuropsychological tests are often performed in the postacute phase to assess recovery and assist in making return-to-play decisions.[7,36,37]

There were 2 aims of the current study: (1) determine whether preinjury RFs are related to the occurrence of concussion and (2) determine whether any postinjury RFs or injury characteristics are associated with clinical recovery, defined for the purposes of this study as days until symptom-free (DUSF) and days until full RTP (DUFRTP).

METHODS

Procedures

This was a prospective cohort study of male and female collegiate athletes participating in a variety of varsity sports as well as club rugby. Data collected before injury or after injury included information about RFs and clinical assessments. Athletes were prospectively followed for the subsequent occurrence of SRC and length of clinical recovery after SRC. The baseline assessments included the SCAT3,[3] Generalized Anxiety Disorder 7-item (GAD-7),[38] Patient Health Questionnaire (PHQ-9),[39] ImPACT,[40] and self-reported history of concussion, ADHD, LD, headaches/migraines, and depression or anxiety. The SCAT3 was administered after injury by the certified athletic training (ATC) or team physician staff as part of the acute assessment for SRC. All athletes diagnosed with SRC, using the definition of the CISG,[41] were evaluated and managed by a team physician. ImPACT was administered within 24 to 48 hours of injury by the ATC or research staff.

Athletes diagnosed with SRC were managed according to the CISG guidelines.[41] After an initial period of brief relative rest (24-48 hours), cognitive (eg, return to classes) and physical activity (eg, walking and biking) were gradually introduced on a symptom-limited basis until full academic and sport participation was achieved. The study was approved by the Institutional Review Panel for Human Subjects Research at Princeton University. All participants provided written consent to participate.

Data Analysis

For aim 1, examining baseline predictors of subsequent concussion, χ^2 analyses were used to explore the association between categorical variables and concussion diagnosis. Logistic regression was used to examine the unique variance associated with previously identified significant predictors of concussion diagnosis. As appropriate, *t*-tests or Mann–Whitney tests were used to examine group differences in continuous variables. For aim 2, Spearman correlations were used to examine the association between variables of interest and DUSF and DUFRTP. Partial Spearman correlations were used to assess the independent influence of behavioral measures. All analyses were conducted using IBM SPSS 24. To reduce the likelihood of familywise error, significance was set at $P < 0.01$.

RESULTS

Aim 1: Predicting Prospective Concussion

We prospectively assessed 1152 athletes (69% male) at baseline and 145 (75% male) with baseline data were subsequently diagnosed with concussion. On average, participants were 19.42 ± 1.35 years old at the time of baseline testing. A complete listing of all sports and their distribution is provided in the legend of Table 1. Mean time between baseline evaluation and injury was 410.45 ± 332.43 days. Self-reported concussion history was available for 1147 athletes. The majority of these athletes (71.8%) reported no history of concussion, 208 athletes (18.1%) reported 1 previous concussion, 74 (6.4%) reported 2 previous concussions, and 38 (3.2%) reported a history of 3 or more concussions (range 3-5). Common self-reported health conditions included headache (6.9%), LD (2.8%), and depression or anxiety (3.0%).

Analyses

Diagnosis of concussion was not significantly associated with sex [36/354 (10.2%) women concussed and 109/798 (13.7%) men concussed, $\chi^2 = 2.71$, $P = 0.10$] or age (mean age at baseline concussed = 19.37 ± 1.12; mean age at baseline not concussed = 19.43 ± 1.38, t (1150) = 0.49, $P = 0.62$) at baseline evaluation. Type of sport played was significantly associated with subsequent diagnosis of concussion. Logistic regression, with concussion status entered as the dependent

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TABLE 1. Descriptive Statistics for Baseline Evaluation Measures

	n	Mean	SD	Median	Min	Max
GAD-7	1080	1.24	2.12	0	0	22
PHQ-9	1081	0.99	1.80	0	0	26
mSCAT Sx tot	1133	1.76	2.81	1	0	22
mSCAT Sx severity	1128	2.68	5.06	1	0	58
mSCAT immediate memory total	1135	14.75	0.57	15	11	15
mSCAT concentration	1030	4.18	0.94	4	1	5
mSCAT orientation	1136	4.93	0.26	5	3	5
mSCAT delayed memory	1030	3.82	1.12	4	0	5
mBESS firm	1149	3.37	3.55	2	0	30
ImPACT visual memory	875	81.43	11.76	83	34	100
ImPACT verbal memory	875	90.62	8.75	93	51	100
ImPACT visual motor	875	44.55	5.65	45.03	26.13	54.50
ImPACT reaction time	875	0.57	0.75	0.55	0.40	0.96
ImPACT Sx score	875	2.41	4.69	0	0	49

Sports represented include football (17%), men's rugby (10%), women's rugby (8.2%), men's lacrosse (7.6%), men's soccer (6.1%), sprint (lightweight) football (5.7%), men's ice hockey (4.7%), wrestling (4.6%), women's soccer (4.3%), baseball (3.7%), women's ice hockey (3.5%), and men's water polo (3.5%). Additional sports included women's water polo (1.7%), men's volleyball (1.5%), women's volleyball (2.7%), women's lacrosse (2.6%), softball (2.2%), men's basketball (2.3%), women's basketball (1.2%), men's track and field (2.5%), and women's track and field (1.5%).

GAD-7, Generalized Anxiety Disorder-7; mSCAT, modified Standardized Concussion Assessment Tool; m-BESS, modified Balance Error Scoring System (firm stances only); PHQ-9, Patient Health Questionnaire-9; Sx, symptom.

variable and indicator dummy coding of sport type as the independent variable, revealed a main effect of sport type on concussion status (Wald = 44.79, $P = 0.002$). Follow-up χ^2 analyses revealed that athletes playing men's ice hockey were significantly more likely to experience a concussion when compared with the rest of the sample [18/54 (33.33%) versus 127/1098 (11.57%), $\chi^2 = 22.16$, $P < 0.001$]. Nonsignificant trends also suggested that athletes playing sprint (lightweight) football [14/66 (21.21%) versus 131/1086 (12.06%), $\chi^2 = 4.73$, $P = 0.03$] and men's rugby [22/115 (19.13%) versus 123/1037 (11.86%), $\chi^2 = 4.97$, $P = 0.03$] were also more likely to subsequently receive a diagnosis of concussion.

Self-reported concussion history was significantly associated with subsequent occurrence of concussion (Table 2) ($\chi^2 = 14.74$, $P = 0.001$). In particular, 10% of athletes with no previous history of concussion were subsequently diagnosed with concussion, whereas 16% of athletes with one self-reported concussion and 21% of athletes with 2 or more self-reported concussions were subsequently diagnosed with concussion during the study period. We conducted logistic regression with diagnosed concussion status entered as the dependent variable; dummy coded sport and concussion history variables were entered as independent variables. Results revealed that both sport type (Wald = 40.29, $P = 0.007$) and concussion history (Wald = 9.91, $P = 0.007$) accounted for unique variance in subsequent concussion diagnosis, indicating that concussion history predicted a subsequent diagnosis of concussion even after considering concussion risk by sport type. A nonsignificant trend indicated that patients who experienced a concussion during the study performed better on the ImPACT verbal memory composite score (92.08 ± 7.79 vs 90.34 ± 8.90, $z = -2.14$, $P = 0.03$). A nonsignificant trend was also found indicating that acutely concussed athletes obtained higher (ie, worse) scores on the m-BESS (4.07 ± 3.93 vs 3.27 ± 3.49, $z = -2.55$, $P = 0.01$). As shown in Table 3, no significant baseline differences emerged between athletes with and without subsequent concussion on the remaining measures (all P values > 0.05). Similarly, no significant relationships were found between acute concussion and baseline report of headache ($\chi^2 = 0.61$, $P = 0.44$), depression or anxiety ($\chi^2 = 0.48$, $P = 0.49$), or LD ($\chi^2 = 0.26$, $P = 0.61$).

Aim 2: Predicting Recovery Time

We included all athletes evaluated for their first study-related concussion within 2 weeks of injury. Over the course of the study, 163 athletes were diagnosed with concussion. A small number (25) of athletes were lost to follow-up, either because they left campus, or their season was over and they did not

TABLE 2. Association Between Self-Reported Concussion History and Subsequent Concussion Diagnosis

Concussion History	No Concussion	Concussion	Total
0	742	85	827
1	174	34	208
≥2	88	24	112
Total	1004	143	1147

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TABLE 3. Descriptive and Nonparametric Statistics Comparing Baseline Test Results for Athletes Who Did and Did Not Sustain a Subsequent Concussion

	Baseline Results No Subsequent Concussion Diagnosis				Baseline Results Subsequent Concussion Diagnosis					
	n	Mean	SD	Mean Rank	n	Mean	SD	Mean Rank	Z	*P*
GAD-7	943	1.22	2.13	534.91	137	1.42	2.02	579.01	−1.68	n.s.
PHQ-9	944	0.99	1.82	537.54	137	1.05	1.67	564.86	−1.07	n.s.
mSCAT Sx tot	990	1.80	2.84	571.02	143	1.51	2.60	539.17	−1.16	n.s.
mSCAT Sx severity	985	2.74	5.15	568.71	143	2.27	4.38	535.49	−1.22	n.s.
mSCAT immediate memory total	996	14.75	0.57	569.05	140	14.75	0.54	560.50	−0.42	n.s.
mSCAT concentration	1000	4.19	0.94	570.78	139	4.17	0.95	564.38	−0.23	n.s.
mSCAT orientation	996	4.93	0.26	569.15	140	4.92	0.27	563.91	−0.40	n.s.
mSCAT delayed memory	892	3.80	1.12	510.29	138	3.94	1.10	549.17	−1.49	n.s.
mBESS firm	1007	3.27	3.49	564.79	140	4.07	3.93	640.27	−2.55	n.s.
ImPACT visual memory	732	81.27	11.79	435.10	143	82.21	11.60	452.84	−0.77	n.s.
ImPACT verbal memory	732	90.34	8.90	429.93	143	92.08	7.80	479.31	−2.14	n.s.
ImPACT visual motor	732	44.58	5.63	439.01	143	44.42	5.76	432.82	−0.27	n.s.
ImPACT reaction time	732	0.57	0.08	441.17	143	0.56	0.07	421.75	−0.84	n.s.
ImPACT Sx score	732	2.46	4.79	439.96	143	2.12	4.16	427.95	−0.57	n.s.

return to their sport, leaving 138 (34 female) athletes who remained in the study until symptom resolution. On average, athletes were 20.41 ± 1.41 years old. The concussed athletes played a variety of sports, as listed in Table 4, with 41 reporting a history of 1 and 23 reporting 2 or more previous concussions. Mean time from concussion to postinjury SCAT3 testing was 0.83 ± 1.79 days (n = 138, range = 0-13 days) and to postinjury ImPACT testing was 2.48 ± 1.58 days (n = 123, range = 0-13 days). Baseline and postinjury descriptive data for the SCAT3 and ImPACT are presented in Table 5. Mean time to symptom-free was 9.84 ± 11.11 days (n = 138, median = 5 days, range = 1-86), and mean time to DUFRTP was 20.21 ± 19.17 days (n = 98, median = 20.21, range = 4-150, Figure 1).

Baseline Predictors

Days until symptom-free was significantly correlated with DUFRTP (r_s = 0.75, P < 0.001). By contrast, neither recovery measure (DUSF or DUFRTP) was significantly associated with baseline age, sex, self-reported concussion history, headache history, screens for depression (PHQ-9) and anxiety (GAD-7), ImPACT performance indices, SCAT3 performance indices, or baseline symptom report (all P values > 0.05).

Postinjury Predictors

Table 6 shows correlations between postinjury evaluations and DUSF/DUFRTP. Follow-up nonparametric partial correlations using the whole sample revealed that ImPACT total symptom score accounted for unique variance in DUSF symptom duration after accounting for variance associated with reaction time and visual motor composites (r_s = 0.51, P < 0.001). A nonsignificant trend indicated that worse performance on the reaction time composite was associated with longer symptom duration after accounting for ImPACT total symptom score (r_s = 0.20, P = 0.03). The relationship between ImPACT visual motor composite performance and symptom duration was no longer significant after accounting for ImPACT total symptom score (r_s = −0.15, P = 0.10). Postinjury DUSF was not significantly associated with other measures listed in Table 6 (all P values > 0.05).

Follow-up nonparametric partial correlations in the whole sample revealed that ImPACT total symptom score accounted for unique variance in DUFRTP after accounting for variance associated with reaction time and visual motor composites (r_s = 0.31, P = 0.004). Similarly, performance on the reaction time (r_s = 0.27, P = 0.01) and visual motor (r_s = −0.30, P = 0.006) composites remained significantly associated with length of DUFRTP after accounting for variance associated with ImPACT total symptom score. Days until full RTP was not significantly associated with other measures listed in Table 5 (all P values > 0.05). Exploratory analyses with individual symptoms revealed that postinjury ImPACT reports of headache (r_s = 0.35, P = 0.001), balance problems (r_s = 0.30, P = 0.004), dizziness (r_s = 0.28, P = 0.008), and difficulty concentrating (r_s = 0.29, P = 0.005) were associated with more DUFRTP.

DISCUSSION

This study evaluated both male and female collegiate athletes participating in a variety of sports and found that only sport type and previous history of self-reported concussion predicted a subsequent concussion. No other RFs (including sex, self-reported history of migraine, LD, depression, or anxiety) were associated with a subsequent diagnosis of concussion. In addition, none of these preinjury RFs were associated with clinical recovery, defined by DUSF and DUFRTP. By contrast, longer clinical recovery was associated with injury-related variables and neuropsychological test variables, including total number of symptoms, symptom severity, and ImPACT reaction time and information processing speed composites.

In terms of predicting the subsequent diagnosis of concussion, we found that men who played ice hockey had higher rates

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TABLE 4. Concussed Athlete Sport Participation

Sport	No. of Concussed Athletes (138)	% of Total Concussed
Football	20	14.4
Men's rugby	19	13.7
Men's ice hockey	16	11.5
Women's rugby	14	7.2
Men's lacrosse	15	8.9
Sprint football	12	8.6
Wrestling	8	5.8
Women's ice hockey	7	5.0
Men's soccer	4	2.9
Field hockey	4	2.9
Men's volleyball	4	2.9
Women's soccer	3	2.1
Baseball	2	1.4
Softball	2	1.4
Men's water polo	2	1.4
Women's lacrosse	1	0.7
Men's basketball	1	0.7
Women's basketball	1	0.7
Men's track/field	1	0.7
Women's diving	1	0.7
Men's crew	1	0.7
Women's volleyball	1	0.7

of concussion than athletes in other sports. Sprint (lightweight) football and men's rugby exhibited a trend toward significance. Interestingly, participation in American football was not associated with an elevated risk for subsequent diagnosis of concussion when compared with other contact/collision sports. This finding might be related to Ivy League—specific football rules that strictly limit the number of contact practices during the competitive season, compared with the NCAA rules for American football, where during the study period, there were no limits to the number of contact practices. By contrast, these same limitations are not in place for men's ice hockey in the Ivy League. None of the other RFs, including sex, self-reported history of LD, migraines/headaches, depression, or anxiety, were associated with subsequent concussion. In addition, none

TABLE 5. Descriptive Statistics for Acutely Concussed Athletes' Baseline and Postinjury Scores

	Baseline Measures						Postinjury Measures						Baseline vs Postinjury	
	n	Mean	SD	Median	Min	Max	n	Mean	SD	Median	Min	Max	Z Value	*P*
GAD-7	117	1.48	2.14	1	0	12								
PHQ-9	117	1.09	1.77	0	0	12								
mSCAT Sx tot	121	1.51	2.64	0	0	18	138	8.77	4.86	8	0	22	−9.01	<0.001
mSCAT Sx severity	121	2.33	4.15	0	0	29	138	19.08	16.57	14.5	0	92	−8.87	<0.001
mSCAT immediate memory	118	14.74	0.56	15	12	15	118	14.48	1.53	15	3	15	−0.98	n.s.
mSCAT concentration	118	4.16	0.94	4	1	5	117	4.14	1.44	4	0	14	−2.78	n.s.
mSCAT orientation	119	4.92	0.27	5	4	5	118	4.70	0.57	5	2	5	−3.56	<0.001
mSCAT delayed memory	117	3.93	1.11	4	0	5	118	3.51	1.48	4	0	5	−3.04	<0.01
mBESS firm	119	4.10	3.90	3	0	20	117	5.94	5.60	5	0	30	−3.40	<0.01
ImPACT visual memory	122	81.93	11.77	82.50	48	100	124	79.99	13.65	81	48	100	−1.51	n.s.
ImPACT verbal memory	122	91.19	10.89	94	11	100	124	88.77	12.98	93	11	100	−2.41	n.s.
ImPACT visual motor	122	44.11	5.82	44.52	26.95	54.5	124	43.24	6.96	44.70	19.23	54.13	−1.21	n.s.
ImPACT reaction time	122	0.56	0.07	0.55	0.41	0.77	124	0.61	0.11	0.58	0.44	1.21	−4.51	<0.001
ImPACT symptom score	122	2.12	4.28	0	0	25	124	9.69	13.00	4	0	68	−5.95	<0.001

To account for poor effort during baseline testing, analyses were conducted with and without baseline ImPACT performance index outliers with no meaningful change in results.
GAD-7, Generalized Anxiety Disorder-7; mSCAT, modified Standardized Concussion Assessment Tool; m-BESS, modified Balance Error Scoring System (firm stances only); PHQ-9, Patient Health Questionnaire-9; Sx, symptom.

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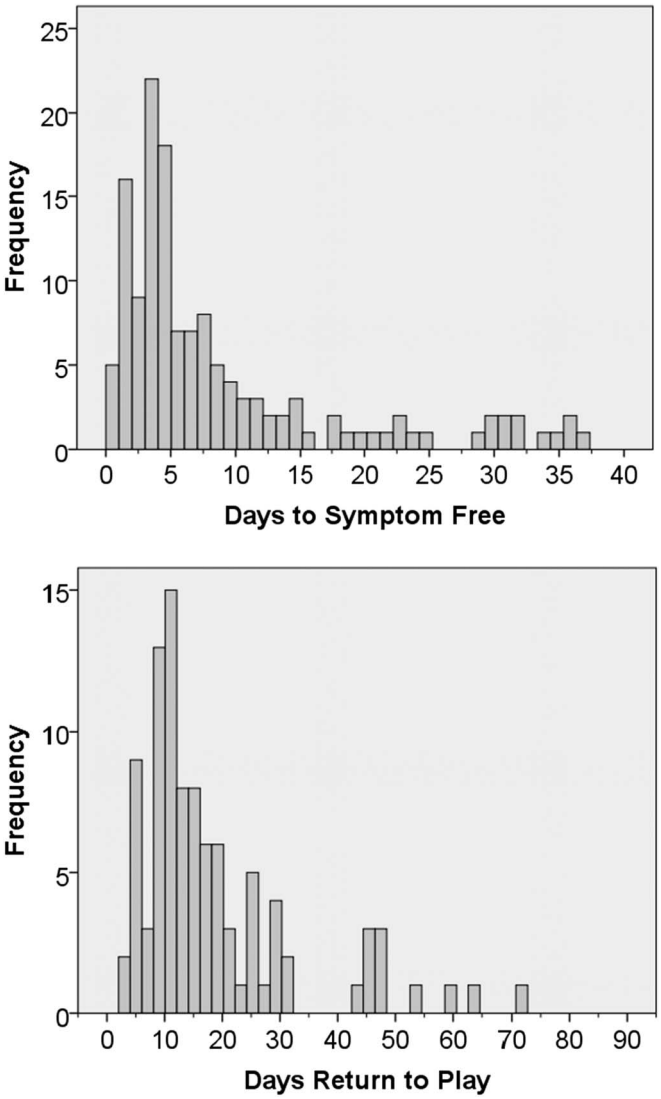

Figure 1. Histograms of days to symptom-free and RTP. To increase readability of figure, one case not shown with days to symptom-free 86 days after injury and one case not shown with RTP 150 days after injury.

of the baseline screening measures, including the SCAT3, PHQ-9, GAD-7, or ImPACT, were associated with subsequent occurrence of concussion.

Our study did not find that any of the preinjury RFs were associated with clinical recovery, which is contrary to the findings of some studies[3,16,20,24,25] but in agreement with many of the conclusions from the most recent systematic review.[10] These researchers raised questions as to whether sex and other RFs with the exception of migraine were associated with prolonged recovery. They also underscored the lack of prospective research examining RFs. The main conclusion from the systematic review was that the most consistent predictor of clinical recovery was the initial number and severity of symptoms.[10]

We found that both symptom severity and total number of symptoms reported after injury were significantly related to prolonged clinical recovery, which has also been reported by others.[4,10,27,29,30,34] Specifically, we found that headache, balance problems, dizziness, and difficulty concentrating were associated with prolonged clinical recovery (DUFRTP). Our findings are in line with previous research that also identify that dizziness and headache at the time of concussion are associated with protracted recovery.[27]

We also found that ImPACT visual motor and ImPACT reaction time composites were significantly associated with DUSF and DUFRTP. The associations with DUFRTP remained significant even after accounting for variance related to ImPACT total symptom score. In supporting the utility of postinjury neuropsychological tests, these findings are in agreement with others[29,31] when evaluating SRC and also in predicting those athletes who may have prolonged recovery.

Although we did not find added benefit in predicting recovery using much of the baseline information that was obtained or baseline ImPACT testing, our study does support recommendations for a multidomain assessment such as the SCAT5 and neuropsychological testing after injury.[7,35–37,42,43] As such, our study provides critical implications not only in terms of assessment of SRC, but also in potentially predicting which athletes may be at risk for prolonged recovery.

A significant strength of our study was an examination of injury RFs prospectively in both male and female athletes who participated in a broad range of sports, thereby minimizing bias and increasing the generalizability of our results. Limitations of our study include the small number of participants that reported a history of headaches/migraines, LD, ADD/ADHD, depression, and anxiety. Another limitation of our study is that we included the PHQ-9 and GAD-7 screening instruments as part of the preseason assessment,

TABLE 6. Correlation Between Postinjury Evaluations and DUSF and DUFRTP

	SCAT3 Total Symptom	SCAT3 Symptom Severity	ImPACT Symptom Score	ImPACT Visual Motor	ImPACT Reaction Time
All evaluations					
DUSF	0.31†	0.33†	0.52†	−0.29*	0.33†
DUFRTP	0.18	0.17	0.32*	−0.33*	0.32*
Acute evaluations					
DUSF	0.26*	0.25*	0.54†	−0.31*	0.40†
DUFRTP	0.20	0.15	0.38*	−0.35*	0.45†

Spearman Rho correlation coefficients. SCAT3 acute evaluations occurred within 1 d of concussion. ImPACT acute evaluations occurred within 2 days of concussion. DUSF and DUFRTP were not significantly associated with ImPACT visual or verbal memory indices.
*$P < 0.01$.
† $P < 0.001$.

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which may be a time of increased stress leading to the potential for bias in reporting. In addition, we also relied on self-report measures for the RFs instead of using verified diagnosis by health care professionals. In addition, we did not take into consideration whether any of the RFs (ie, headaches, migraines, LD, ADD/ADHD, depression, or anxiety) were subsequently treated, which may have altered the results. Additional follow-up time points would strengthen the study. This study only examined Ivy League athletes, which might limit the generalizability to other non–Ivy League collegiate athletes and also to other age groups. Finally, there were a small number of female athletes who sustained concussion, which may have reduced the statistical power to find sex-based differences.

CONCLUSIONS

This study is a step forward in our understanding of which factors predict who may be at risk for a diagnosis of concussion or the length of recovery following SRC among Ivy League athletes. We found that, despite a broad assessment of preinjury RFs, only self-reported concussion history and playing ice hockey were significantly associated with a subsequent diagnosis of SRC. No other RFs or baseline measures predicted clinical recovery, defined as DUSF or DUFRTP. Similarly, our findings also underscore the important role of specific postinjury measures, including symptom number and severity, and select clinical measures, including components of the neuropsychological test battery in predicting longer clinical recovery.

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